Project 1

Step	Environment	Active Hours (per member)	Reproducibility Notes / Alternatives
1. Target structure retrieval	Colab	3–4 h	Use Biopython PDBList in Colab; pin PDB accessions and resolution cutoff in `targets.csv` committed to GitHub. Re-running yields identical outputs.
2. Compound library curation	Colab	6–8 h	Use RDKit Ro5, Veber, and PAINS filters in Colab. Download from NANPDB, AfroDb, and ChEMBL via API; export canonical SMILES to `library_final.csv`. Save raw downloads because databases update without versioning.
3. 3D conformer generation	Colab	2–3 h	Use RDKit `AllChem.EmbedMolecule` + MMFF94 with `randomSeed=42`. Same member/session as curation. Seed-fixed runs are reproducible.
4. Receptor preparation	Colab + Web	4–6 h	Use CHARMM-GUI for protonation and Meeko for PDBQT generation. Free alternative to Schrödinger Prep

			Wizard is Meeko + PDB2PQR web.
5. Docking validation (re-docking)	Colab	3–4 h	Use AutoDock Vina Python API and RMSD via MDAnalysis. Set `seed=42` and commit config file. Avoid PyRx GUI because it is not scriptable.
6. High-throughput docking	Colab + Lightning.ai	5–8 h	Use batch Vina loops; split libraries into chunks. Lightning.ai is better for >5,000 compounds. AutoDock-GPU is a faster open-source alternative. Commit receptor PDBQT, grid box parameters, config, and results CSV.
7. Hit filtering & ADMET	Colab + Web	4–5 h	Use SwissADME, pkCSM, and ADMETlab 2.0. Cache raw API outputs. DeepPurpose or ADMET-AI are good offline alternatives.
8. MD system setup (CHARMM-GUI)	Web	4–6 h	Use CHARMM-GUI Solution Builder to generate GROMACS-ready files. Commit all output files to GitHub. Avoid manual editing of output files unless

			scripted and documented.
9. MD minimisation + equilibration	HPC / VM	3–4 h	Use GROMACS SLURM scripts and version-tagged Docker images. Not practical on Colab because session resets can corrupt checkpoints.
10. MD production (100 ns)	HPC / VM	6–10 h active	Run one SLURM job per complex with checkpointing every 10 ns. Prioritize top hits first. This is the main bottleneck and requires GPU-backed HPC or cloud VMs.
11. Enhanced MD / metadynamics	HPC / VM	6–8 h active	Use GROMACS + PLUMED in Singularity or BioContainers. Commit `plumed.dat` to GitHub. GaMD via NAMD3 is a fallback if PLUMED is unavailable.
12. MM-PBSA binding free energy	Lightning.ai	4–6 h	Use `gmx_MMPBSA` on Lightning.ai after transferring trajectories from HPC. Lightning is preferred over Colab because installs persist.
13. Trajectory analysis & visualisation	Colab	8–12 h	Use MDAnalysis, PLIP, matplotlib, seaborn, and

			Py3Dmol. Generate figures from committed summary CSVs rather than raw trajectories.
14. Final figures & manuscript package	Colab	6–10 h	Use a single `manuscript_figures.ipynb`, plus `requirements.txt` and `environment.yml`. Mint a Zenodo DOI from the GitHub release.

Project 1 totals

- Total active hours across all members: ~64–94 h
- Critical path: MD production + enhanced MD
- HPC-only steps: 9–11

Galaxy Europe as a supplementary option for Project 1

Area	What Galaxy Europe Can Do	Limitation
Compound filtering	RDKit-based filtering tools are available	Good for early-stage library refinement only
Docking	Galaxy has an AutoDock Vina wrapper	Fine for docking and redocking, but not for large-scale downstream MD
ADMET	Some web-integrated ADMET tools exist	Limited compared with dedicated offline workflows
MD post-processing	Basic MDAnalysis-style post-processing is possible	Limited for large trajectories and advanced workflows
Production MD	Limited GROMACS wrappers exist	Not suitable for long 100 ns production runs or meaningful GPU compute
PLUMED / Metadynamics	Not available	No Galaxy wrapper exists

gmx_MMPBSA	Not available	Must still be run on Lightning.ai or local/HPC compute
Large trajectory handling	Browser-based upload possible	10–50 GB trajectory files are impractical to manage through Galaxy

So Galaxy Europe covers roughly the first 40% of the Project 1 workflow well, especially compound filtering, docking, and some ADMET work. After that, the workflow hits a hard wall at the computationally expensive MD stages. For your team, Galaxy is best used to offload docking and early screening so that Oracle Cloud or Google Cloud credits can be reserved specifically for the MD-intensive parts.

Project 3

Step	Environment	Active Hours (per member)	Reproducibility Notes / Alternatives
1. Data access & cohort download	HPC / VM	3–5 h	Use GDC Transfer Tool CLI to download to cloud or mounted storage. Commit `gdc_manifest.txt` for exact reproducibility. Raw BAMs should stay on cloud or HPC storage.
2. Raw QC (FastQC / MultiQC)	HPC / VM	2–3 h	Run FastQC as a SLURM array and aggregate results with MultiQC. Commit the MultiQC HTML report to GitHub.
3. Trimming (Trimmomatic)	HPC / VM	2–3 h	Commit adapter FASTA files and parameter strings. Best reproducibility comes from wrapping

			this in `nf-core/rnaseq`.
4. Alignment (STAR / BWA-MEM2)	HPC / VM	4–6 h active	Commit GRCh38 and GENCODE v43 reference checksums. Alignment is one of the true bottlenecks for large cohorts.
5. Duplicate marking + BQSR	HPC / VM	3–4 h active	Use version-tagged GATK Docker images and commit the Docker command. Shared GATK resources should be stored once for the team.
6. Somatic variant calling (Mutect2)	HPC / VM	5–8 h active	Use SLURM arrays for paired tumour-normal calling. Commit PoN VCFs and log exact settings. Pre-called GDC MAFs can bypass this step entirely.
7. CNV calling (CNVkit)	HPC / VM + Lightning.ai	3–4 h	Run CNVkit in batch mode and export `.cns` / `.cnr` outputs. GISTIC2 can be run later through GenePattern.
8. VEP annotation	Colab + Web	3–4 h	Use Ensembl VEP REST API and cache raw JSON outputs. Record the Ensembl release version used.

9. Mutational signatures (SigProfiler)	Lightning.ai	3–5 h	Use SigProfilerExtractor in Lightning.ai because NMF can take hours. Commit summary outputs and package versions.
10. RNA-seq quantification & normalisation	Colab	3–4 h	Use TCGAbiolinks to retrieve GDC STAR counts and DESeq2 for normalisation. Record the GDC release version.
11. Differential expression (DESeq2)	Colab	5–7 h	Use `set.seed(42)` and commit the design formula, contrast, and shrinkage method. Export full DEG tables.
12. Pathway enrichment (clusterProfiler / GSEA)	Colab	3–4 h	Use clusterProfiler, fgsea, and version-pinned msigdbr. Commit all enrichment result tables.
13. Multi-omics integration (MOFA2)	Lightning.ai	5–7 h	Use mofapy2 with mutation, CNV, and expression matrices. Save trained models as `.hdf5` for later use.
14. Survival analysis (Cox / KM)	Colab	4–5 h	Use survival, survminer, and rms packages with TCGA clinical data. Entirely tabular and does not require HPC.

15. WGCNA co-expression	Lightning.ai	3–4 h	Use WGCNA with `set.seed(42)` and commit hub genes, module membership, and soft-threshold plots.
16. Genomic visualisation (maftools, OncoPrint)	Colab	5–7 h	Use maftools, ComplexHeatmap, and circlize. Record BiocManager version and commit MAF files.
17. Target prioritisation	Colab + Web	3–4 h	Use DGIdb, Open Targets, OncoKB, and CancerVar. Cache API outputs and log version numbers because databases update frequently.
18. Final figures & manuscript package	Colab	8–12 h	Commit all intermediate files such as MAFs, count matrices, CNV segments, and MOFA2 models. Use Zenodo DOI and environment pinning for reproducibility.

Project 3 totals

- Total active hours across all members: ~67–96 h
- Critical path: Alignment + variant calling
- HPC-only steps: 1–7
- From step 8 onward, the project can be run almost entirely on Colab and Lightning.ai

Galaxy Europe for Project 3

Area	What Galaxy Europe Can Do	Limitation

Raw QC	FastQC and MultiQC are fully available	Works very well with no major limitations
Trimming	Trimmomatic and Cutadapt wrappers are available	Suitable for the full trimming stage
Alignment	STAR, HISAT2, and BWA wrappers exist	Slower than dedicated HPC but still practical for moderate datasets
Variant calling	GATK and Mutect2 wrappers are available	Large cohorts may still be slow, but the workflow is feasible
CNV analysis	CNVkit and related tools can run	Fine for standard cohort sizes
VEP annotation	VEP wrappers exist	Straightforward and reproducible
Expression analysis	DESeq2, edgeR, and limma are available	Strong support for downstream expression analysis
Pathway analysis	GSEA and enrichment workflows are supported	Fully adequate
Visualisation	OncoPrint-like visualisation and heatmaps are available	Some advanced custom plots may still be easier in R or Colab
Workflow sharing	Galaxy workflows can be exported and shared as `.ga` files	Strong reproducibility advantage

For Project 3, Galaxy Europe can support almost the entire workflow end-to-end, especially if you use pre-called MAFs and preprocessed count matrices. Unlike Project 1, there is no unavoidable HPC wall in the middle of the analysis. The only real limitation is that very large alignment or variant-calling jobs may be slower than on a dedicated HPC cluster, but they are still practical. That is why Project 3 is the safer option for a team without guaranteed HPC access.